CivicConnect

Project Engineering Document v3.0

Construction, verification and release evidence index

MERGED - 17 OCTOBER 2026 (commit c7e92de8f818b66ca1ecc0faf88b0987c4b17b40)
M3 evaluation-slice status: APPROVED by Don and Masego, merged to main. Release decision: NOT READY for final baseline/public production release - these are separate questions; see the residual-risk table.
Prepared for Emile, Don and Masego.

This continues the repository PED: M1 sections 01-15 and M2 sections 16/17/19/20/21 remain the historical sources. This export indexes those sources and adds section 22; historical sign-offs retain their original scope.

Candidate evidence: 60 automated tests passed; two live HTTP journeys; p95 0.422 seconds over five minutes with 10 users/500 requests; local restart and backup/restore passed. Evidence boundaries and remaining gaps follow.

22. M3 construction, verification and release review

CivicConnect - PED v3.0, merged 16-17 October 2026. Prepared by Emile with AI assistance recorded in AI-036 and AI-038/042. Continues the existing M1/M2 PED sources; no historical signature or baseline status is changed.

22.1 Candidate and construction

Merged into `main` **at commit** `c7e92de8f818b66ca1ecc0faf88b0987c4b17b40` **via PR #85**, with real, independently-confirmed approvals from both Don and Masego against that exact commit. PR #85 folded together the backend routes, route tests, UI templates and dependency-check changes originally opened as PRs #80-84, plus PR #86 (the full role x operation authorisation matrix across all four roles, and the RTM scope-wording corrections both Don and Emile made after independent AI-assisted review). PRs #80, #82, #83 and #84 were closed as superseded once their content was verified present on `main` - not merged separately, to avoid duplicate/conflicting code. PR #81 (dependency-audit workflow) was independent of this chain and merged separately. Source revisions and file hashes are indexed in `docs/release/evidence/2026-10-16/source-manifest.json`. The registration addition is identified separately by `registration-source-manifest.json`.

Implemented slice: public Requester registration, login, submission, own list/detail/status; Staff login, queue/status filter/detail/acceptance; acceptance notification through in-app and SMTP; Manager/Administrator login with a safe landing page. `docs/deployment/m3-interface-agreement.md` records actual routes, forms, roles and error behaviour. `CHG-002` records the proposed release-slice exclusions and impact without rewriting baselined priorities.

Two integration defects were corrected: missing CSRF tokens in shared logout forms on list/landing pages (DEF-004), and a mismatched audit-actor field in the detail template (DEF-005). Both have regression checks.

22.2 Architecture and configuration conformance

The candidate follows layered server-rendered Flask, service-level rules, SQLite persistence, transactional state/audit/outbox changes and an in-process notification worker. ADR-002/003/004/006 remain Accepted in the source; ADR-001/007 remain Proposed. DEC-008 remains Deferred. **The six Issue #65 decisions are now logged in** `docs/decisions/decision-log.csv` **as DEC-013 through DEC-018** (§10.4), reconciled directly against each ADR's actual current status - Deferred for DEC-013/018 (ADR-001/007, still Proposed), Decided for DEC-014/015/016/017 (ADR-002/003/004/006, Accepted). Logging them does not itself move ADR-001 or ADR-007 to Accepted; that is a separate, still-open team decision. Issue #65 is closed - the gap it raised (ADRs existing without a corresponding register entry) is fixed, not the underlying architecture/auth placement decisions themselves.

Staging is a distinct loopback Waitress process with a generated synthetic database, separate credentials and local STARTTLS SMTP sink. Config/secrets and the database stay Git-ignored. This simulates staging and does not establish public HTTPS, availability or production durability. See `docs/deployment/m3-staging.md`.

22.3 Verification results

Evidence	Result	Scope
pytest automated suite	60 passed (re-verified fresh in an isolated worktree, not just the commit message)	39 historical + 11 registration cases (registration-regression.xml) + 10 new role x operation matrix cases (PR #86)
Backend regression follow-up	16 passed	Includes corrected rendered audit actor assertion
Role x operation matrix	10/10 passed, including the previously-missing positive case (assigned Staff viewing their own accepted request)	`tests/test_m3_role_operation_matrix.py`; `docs/quality/role-operation-matrix.md`; closes the coverage gap Don's AI-assisted review found on the first integration commit
Ruff / Bandit / dependency audit	Clean / no medium-high findings / no known application dependency findings	Tool results at check time, not complete security proof
Live HTTP system journey 1	Passed	Requester login, submit #501, receipt, list/detail, logout and cross-requester denial
Live HTTP system journey 2	Passed	Staff filter/detail/accept, requester status, in-app notification and local SMTP capture
Browser walkthrough	Request #502 submitted and accepted	Actual browser UI operated; form/receipt screenshots retained; no checked-in browser automation claim
NFR-001 five-minute load	8,503 successful; 0 failures; p95 0.422s	10 distinct Staff users, 500 filtered records, local warm instance, target <=2s
Local process restart	Passed	Accepted #501 retrieved after stop/start of same-file staging
Online backup/separate restore	Passed	Integrity check OK; request/audit/outbox counts preserved

`docs/release/evidence/2026-10-16/` holds execution reports. The five-second smoke run is explicitly insufficient for NFR-001. The subsequent 300-second run supplies the stated measurement. The load/restart/backup evidence predates registration and was not rerun. Registration-live.json separately confirms public signup and login; email verification and signup rate limiting remain outstanding. No browser rendering benchmark or production-capacity claim is made.

22.4 Traceability and quality controls

`docs/requirements/RTM.csv` adds an execution-evidence column for the implemented slice without promoting baseline statuses. `docs/quality/test-inventory.md` remains the historical kickoff; `docs/quality/m3-verification-handoff.md` records the new boundary and remaining independent checks. Requirements, acceptance criteria, code, named tests, execution reports, DEF-004/005 and release conditions are connected through these artefacts.

GitHub workflows provide automated tests, static analysis, dependency audit and secret-pattern checks. Local results are not GitHub CI results. Current CI and branch-protection enforcement must be checked on the signed final revision. Two separate non-author students must review it; two accounts belonging to one person do not create two reviews.

22.5 Residual risk and release decision

NOT READY for final baseline/public production release. The scoped local demonstration candidate has passing evidence, while independent reviews, formal scope/decision reconciliation and the final committed revision remain pending. See the full release summary for conditions and owners.

Unresolved claims include durable hosted storage, external SMTP/TLS, dormant legacy-hash disposition, complete role x operation coverage, stuck/duplicate outbox recovery, reversible schema/code rollback, sustained availability, cold-start, capacity and accessibility. A local backup drill does not prove daily backup scheduling or retention. Local restart success does not close the M2 Render redeploy failure.

Render documents that free-service local files disappear on restart/redeploy/spindown, which is consistent with the observed M2 failure. The historical instance configuration was not independently recovered. Avoiding redeploy alone cannot ensure durability. PostgreSQL remains a candidate, not an adopted platform.

22.6 Evidence index and individual defence

Engineering concern	Authoritative artefact
M1 business/scope/requirements baseline	PED sections 01-15 and docs/requirements/ registers
M2 architecture/data/patterns/integration/baseline	PED sections 16, 17, 19, 20, 21 and docs/decisions/adr/
Current route/UI contract	docs/deployment/m3-interface-agreement.md
Controlled staging and recovery	docs/deployment/m3-staging.md; scripts/m3_staging.py; scripts/m3_verify_recovery.py
Test classifications and remaining matrix	docs/quality/m3-verification-handoff.md; tests/
Performance scenario and result	performance/load_test_staff_queue.py; performance.json
Live journeys and restart evidence	scripts/m3_verify_live.py; live-system-journeys.json; process-restart.json
Traceability and defects	docs/requirements/RTM.csv; docs/quality/defect-register.csv
Debt and residual risks	docs/quality/technical-debt-register.md; docs/release/m3-release-evidence-summary.md
Decisions/scope requiring review	docs/release/team-decision-review.md; docs/decisions/decision-log.csv (DEC-013..018); CHG-002
AI accountability and progressive history	docs/governance/ai-usage-register.csv; PROJECT_HISTORY.md; final PR/commit

Each student should be able to demonstrate: FR-014/AC-014 -> service version guard -> acceptance regression -> live journey #501 -> audit/outbox/notification evidence -> residual outbox recovery risk -> release condition. Emile should also explain NFR-001's dataset, concurrency, warm-up, duration, p95 method and local-only limits.

22.7 Final review actions - status as of 17 October 2026

1. ~~Emile inspects this candidate~~ - done, repeatedly, across each corrected commit, with real re-execution each time (not just re-reading commit messages).

2. ~~Don reviews backend/migration/outbox, existing ADR statuses and the six draft register rows~~ - done. Don's AI-assisted review found real gaps across two rounds (stale test counts, a coverage overclaim, inaccurate test-layer history, a scope-approval wording overclaim, then a second stale claim after the first fix); all were corrected and re-verified.

3. ~~Masego's own CHG-002 decision/date~~ - done, recorded as a PR #86 comment, confirmed via the GitHub API.

4. ~~Branch-protection ruleset requiring an unconfigured Copilot review~~ - done. Don removed the `copilot_code_review` condition from `main-protection-basic-02` directly (confirmed via the GitHub API before and after) rather than leaving the rule in place and relying on a bypass.

5. ~~2 real non-author approvals on the final commit~~ - done. Don and Masego both approved PR #85 against commit `e76b96e74515e5cc813c6fba4cd1805362e5c020`, with all 7 CI checks green and no conflicts.

6. ~~Merge~~ - done. PR #85 merged to `main` at `c7e92de8f818b66ca1ecc0faf88b0987c4b17b40`. PR #81 merged separately. PRs #80/82/83/84 closed as superseded, each verified present on `main` first.

7. ~~Six Issue #65 decisions logged~~ - done, as DEC-013..018 in `decision-log.csv` and PED §10.4. Issue #65 closed.

Still genuinely open, not closed by any of the above: ADR-001 and ADR-007 remain Proposed - merging to `main` logs the decisions, it does not accept them. Durable hosted storage, external SMTP/TLS, dormant legacy password hashes, sustained availability, capacity and accessibility all remain unverified - see `docs/release/m3-release-evidence-summary.md`'s residual-risk table. This milestone's evaluation slice is complete and merged; a production release is a separate, later decision.

Sources checked 16 October 2026: https://flask.palletsprojects.com/en/stable/testing/ and https://render.com/docs/free. Repository artefacts above supply implementation and execution evidence.